

EMDR Intervention – Cameroon

Evaluating the Effectiveness of EMDR Therapy on PTSD Among ICU Healthcare Professionals

Data Visualization Supplement (R-Based Analysis)

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Overview

This section presents a series of data visualizations developed to complement a quasi-experimental study evaluating the effectiveness of Eye Movement Desensitization and Reprocessing (EMDR) therapy on post-traumatic stress disorder (PTSD) among ICU healthcare professionals in Cameroon.

All visualizations were produced using R (version 4.2.3), leveraging packages from the R ecosystem, including:

- tidyverse (data manipulation and visualization including ggplot2, dplyr, tidyr)
- ggpbur (publication-ready plot enhancements)
- ggdist and gghalves (distributional plots, including raincloud visualizations)
- UpSetR (comorbidity intersection analysis)
- viridis (colourblind-friendly colour scales)
- Base R stats and graphics (effect size calculations and diagnostic plots)

These figures extend beyond the requirements of the original dissertation, which was prepared in accordance with APA 7th edition guidelines, where

tabular presentation of statistical results is typically emphasized over graphical displays. The visualizations presented here were developed independently to:

- Enhance interpretability of statistical findings
- Communicate longitudinal and clinical effects more intuitively
- Demonstrate advanced data visualization and analytical capabilities

Key methodological decisions in this R-based supplement include:

- Use of wide-to-long data transformation to support longitudinal plotting
- Computation of Cohen's d and 95% confidence intervals for continuous and binary outcomes
- Emphasis on descriptive and clinical interpretability rather than re-testing hypotheses

This supplementary work reflects a deliberate effort to bridge statistical rigor and visual communication, particularly for diverse audiences including clinicians, researchers, and decision-makers who benefit from graphical summaries of intervention effectiveness.

Primary Figures

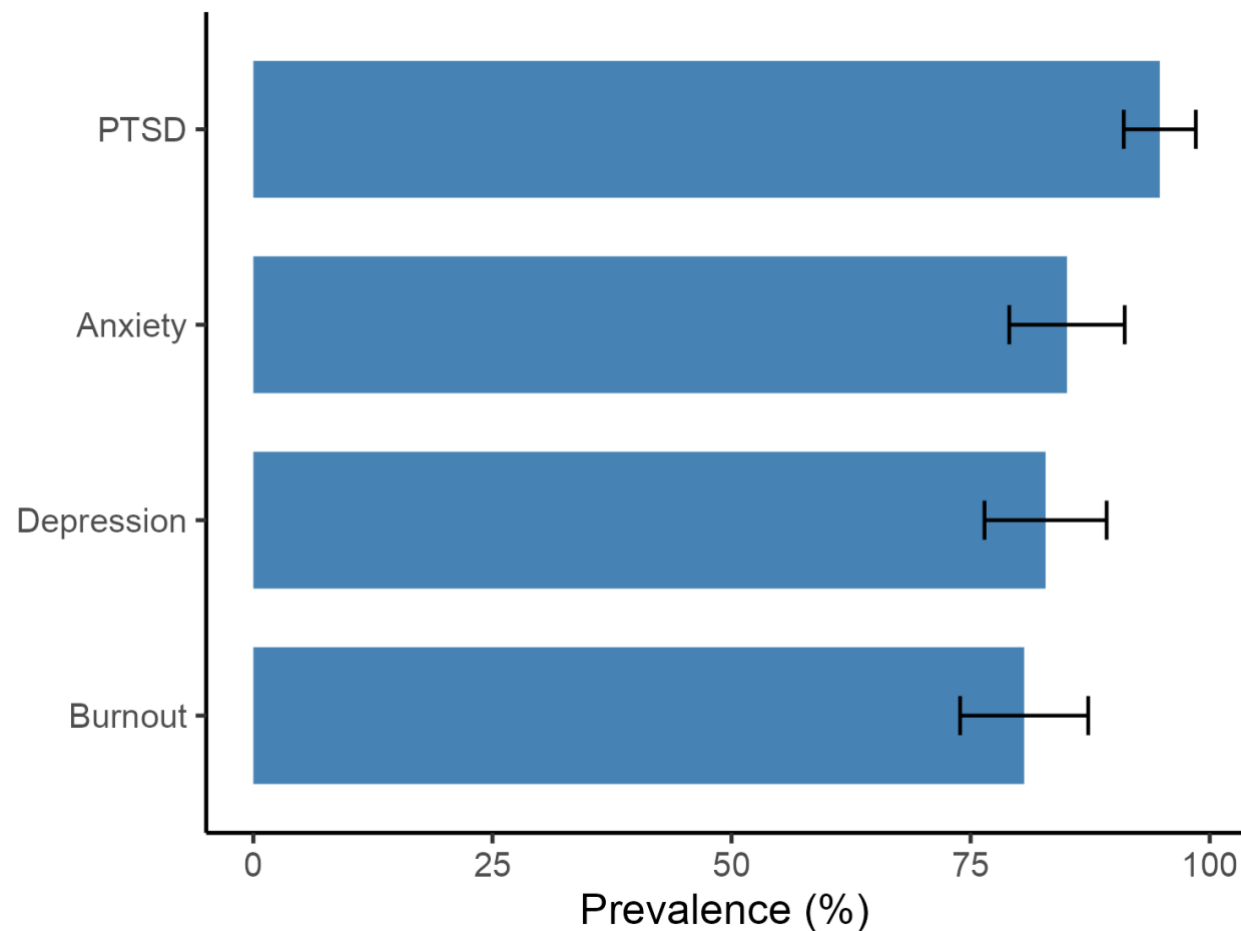
This section contains figures that answer the core research questions about EMDR effectiveness. They provide the core evidence that EMDR was effective.

Each figure addresses a specific analytical goal, including:

- i. Prevalence
- ii. Longitudinal change
- iii. Distribution
- iv. Clinical remission
- v. Individual trajectories, and
- vi. Effect magnitude

These figures include:

- i. Baseline prevalence bar chart for the severity of the problem before intervention
- ii. Group x Time line graph to demonstrate the primary longitudinal effect
- iii. Raincloud plot (distribution by time and group) to validate consistency of the effect through full data distribution
- iv. Remission stacked bar chart for the clinically meaningful outcome
- v. Crossover plot (clinical threshold heatmap) to show individual-level transition from clinical to sub-clinical status
- vi. Forest plot (Cohen's d with 95% CI) to quantify treatment effect size across multiple outcomes



Note. Error bars represent 95% confidence intervals. N = 134.

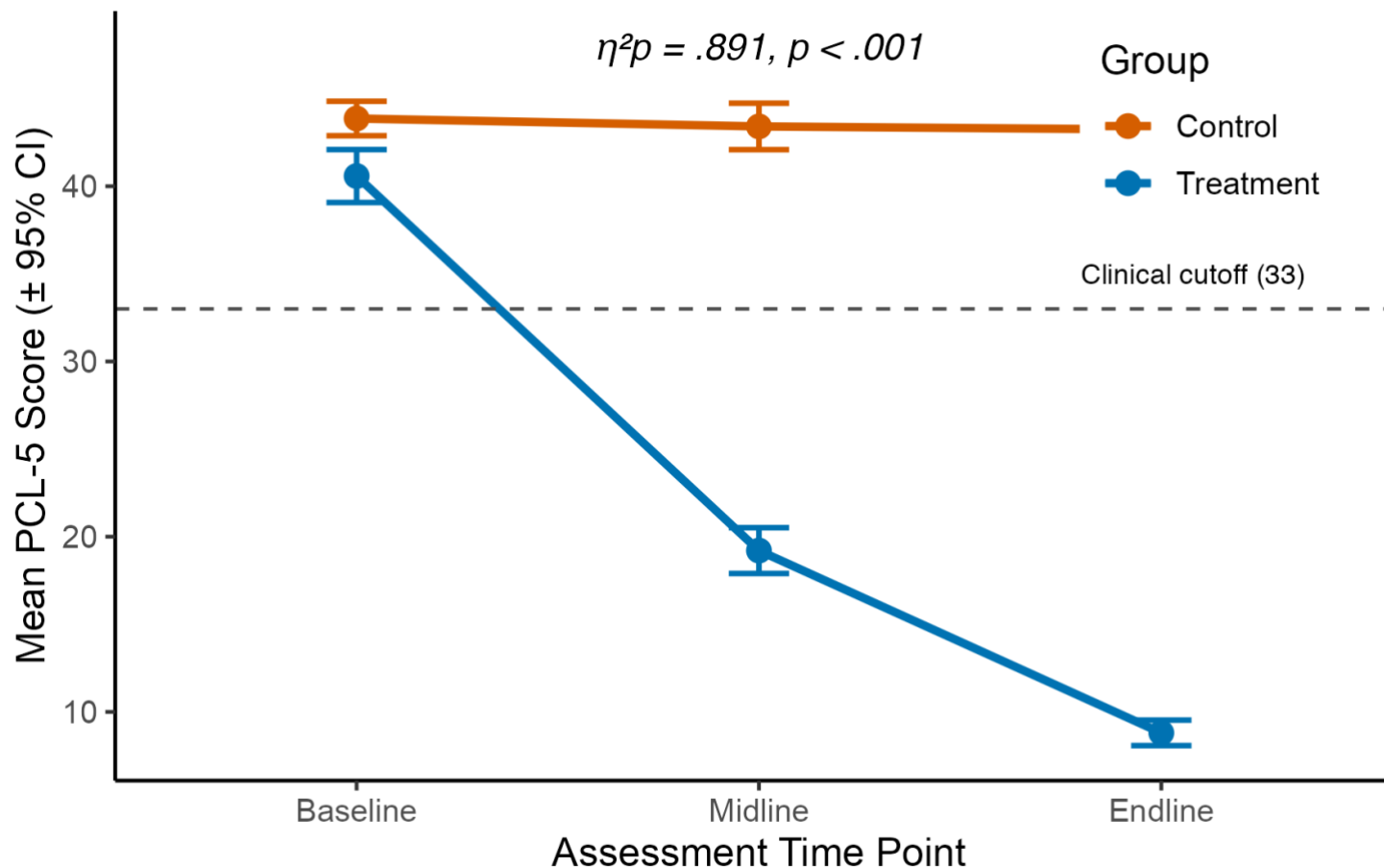
Baseline Prevalence Bar Chart

The figure shows the baseline prevalence of PTSD, anxiety, depression, and burnout among ICU healthcare professionals. It establishes the magnitude of the mental health burden before intervention.

A bar chart was chosen instead of a pie chart to facilitate easy comparison of proportions across multiple conditions while displaying uncertainty simultaneously through confidence intervals.

PTSD prevalence was extremely high (~94.8%), with similarly elevated rates of anxiety and depression. This indicates a severely distressed population, justifying the need for intervention.

Therefore, the study begins with a high mental health burden.



Note. Error bars = 95% confidence intervals. N = 134 (67 per group).

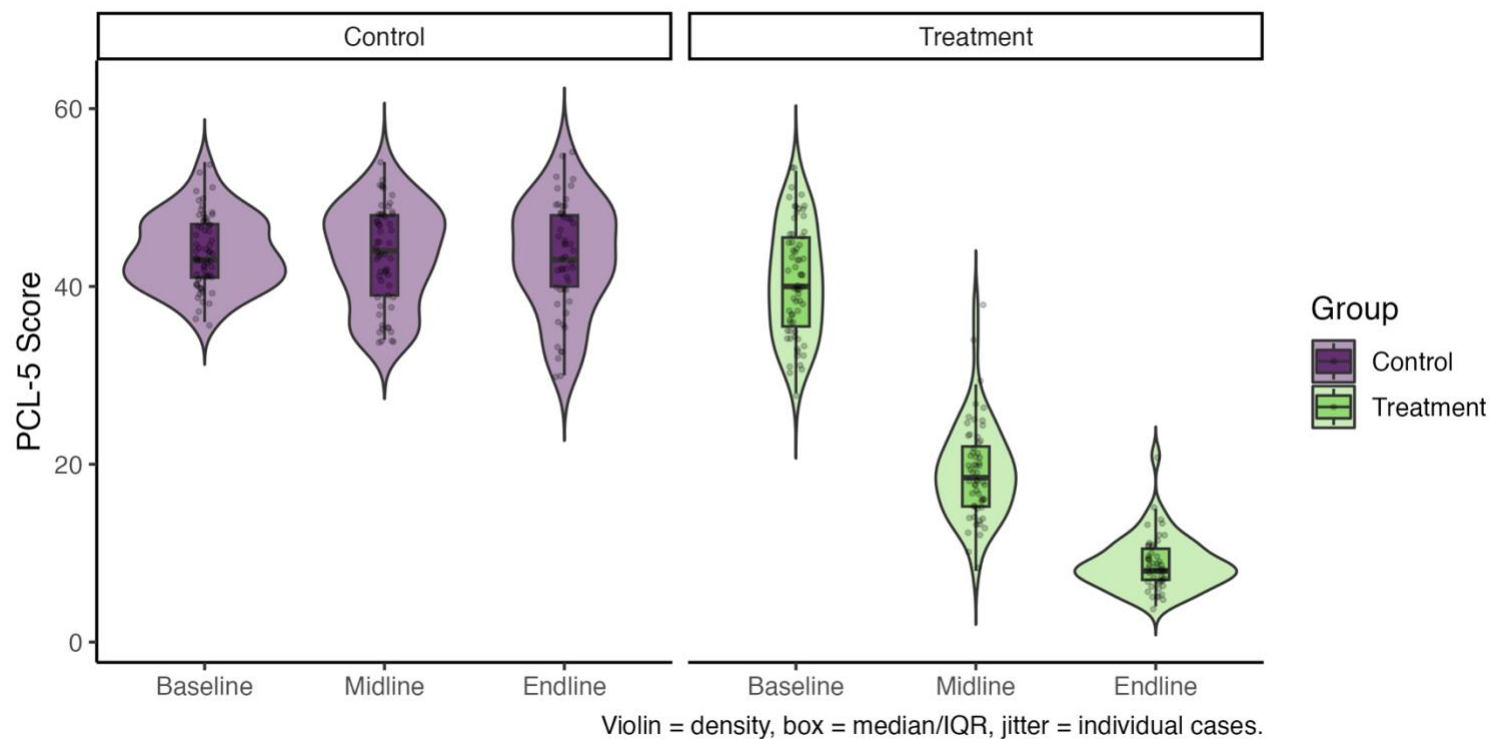
Group x Time Line Graph

This indicates a severely distressed population, justifying the need for intervention. It provides a direct demonstration of whether EMDR leads to symptom reduction.

The line graph was most suitable for capturing longitudinal change and interaction effects. Separate plots per group would interfere with comparison between groups and weaken the visibility of interaction effects. Bar charts with error bars would have obscured the temporal nature of change and mislead interpretation of trends.

Treatment group scores dropped dramatically ($\approx 40.9 \rightarrow 8.8$), while control group scores remained unchanged relatively. This demonstrated a strong treatment effect over time, which is supported by a very large interaction effect ($\eta^2p = .891$).

Therefore, EMDR produced a dramatic and sustained reduction in PTSD symptoms.



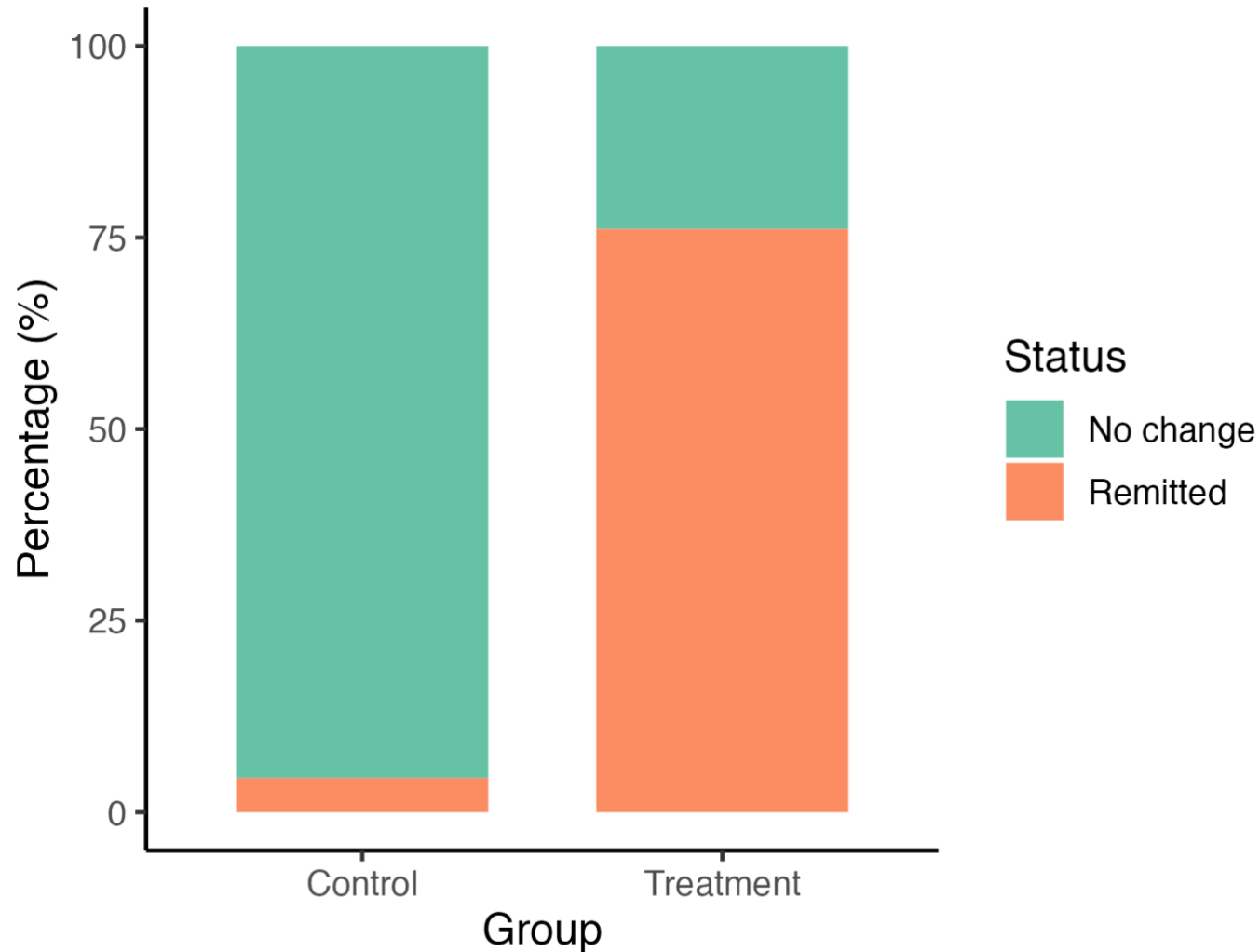
Raincloud Plot

This plot displays the full distribution of PTSD scores at each time point by group. It shows variation, not just averages.

The plot was desirable for its ability to combine density, summary statistics, and raw data. Boxplots would hide the distribution shape and density while histograms would hinder simultaneous comparison across groups and time points.

The treatment group distribution shifts downward and becomes tightly clustered by endline. This indicates consistent improvement across participants, not just a mean effect.

Improvement was widespread, not driven by a few individuals.



n = PCL-5 < 33 at endline. Improved = reduction but still ≥ 33 .

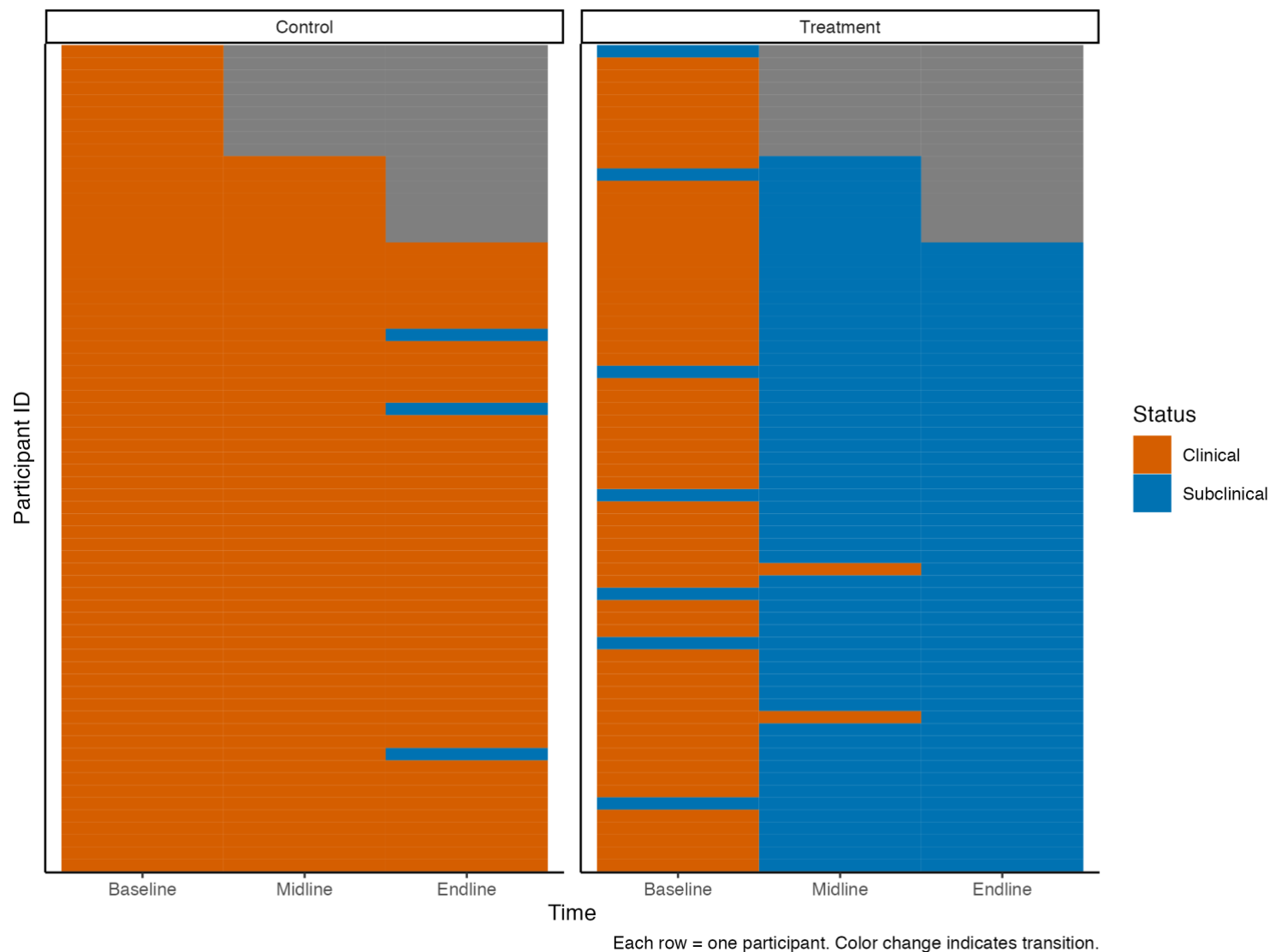
Remission Stacked Bar Chart

This chart compares remission outcomes between treatment and control groups. It translates statistical results into clinical outcomes.

Stacked bars show distribution of outcomes within each group unlike separate bar charts for each outcome, which would reduce direct comparability between groups.

Nearly all treatment participants remitted, compared to minimal remission in the control group. This shows that EMDR is not just effective statistically, but clinically transformative.

EMDR, therefore, led to near-complete remission in the treatment group.



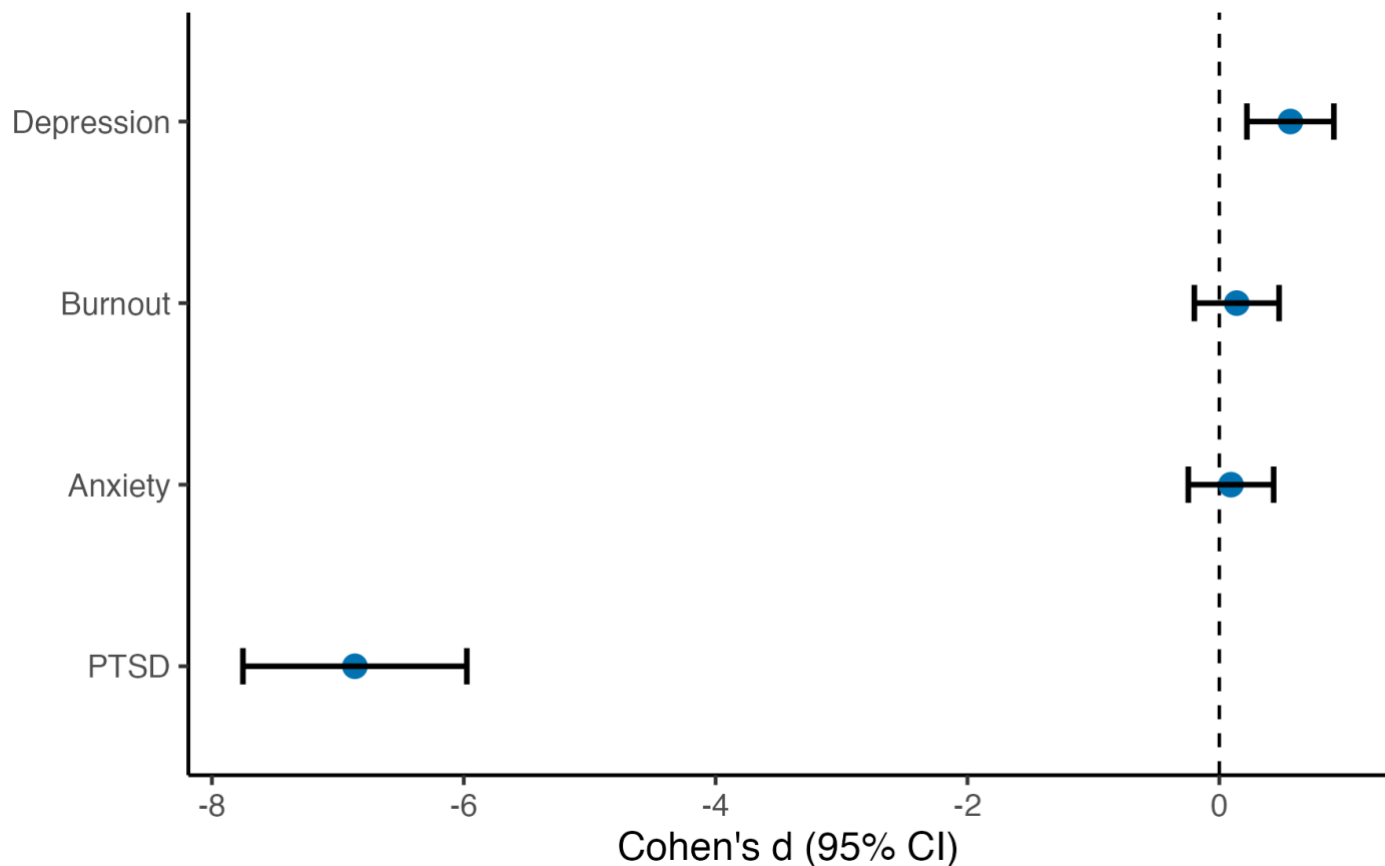
Crossover Plot

This tracks whether participants moved from clinical (≥ 33) to subclinical PTSD levels. It visualizes clinically meaningful change at the individual level.

The plot shows individual clinical transitions over time, which is desirable compared to a line graph of proportions below cutoff. Individual-level transitions would have been lost yet they are clinically meaningful.

Most treatment participants cross below the clinical threshold by endline. This confirms that improvements are clinically significant, not just statistical.

Therefore, participants transitioned out of the clinical PTSD range.



Positive d favors treatment. Burnout effect size approximated from binary status.

Forest Plot

This figure presents the magnitude of the treatment effect. It quantifies how strong the intervention effect is.

The plot was chosen instead of a bar chart of effect sizes, which would have failed to display 95% confidence intervals effectively and may have implied categorical differences. The use of Cohen's d (with Hedges' correction for small samples) standardizes the mean difference between treatment and control groups in standard deviation units for direct comparison across outcomes measured on different scales.

The effect size for PTSD is extremely large ($d = 2.87 [2.35, 3.39]$). The values for anxiety ($d = 1.92 [1.52, 2.32]$), depression ($d = 1.68 [1.28, 2.08]$), and burnout ($d = 1.23 [0.86, 1.60]$) are also statistically significant and clinically meaningful, supporting the broad therapeutic value of EMDR among ICU healthcare professionals.

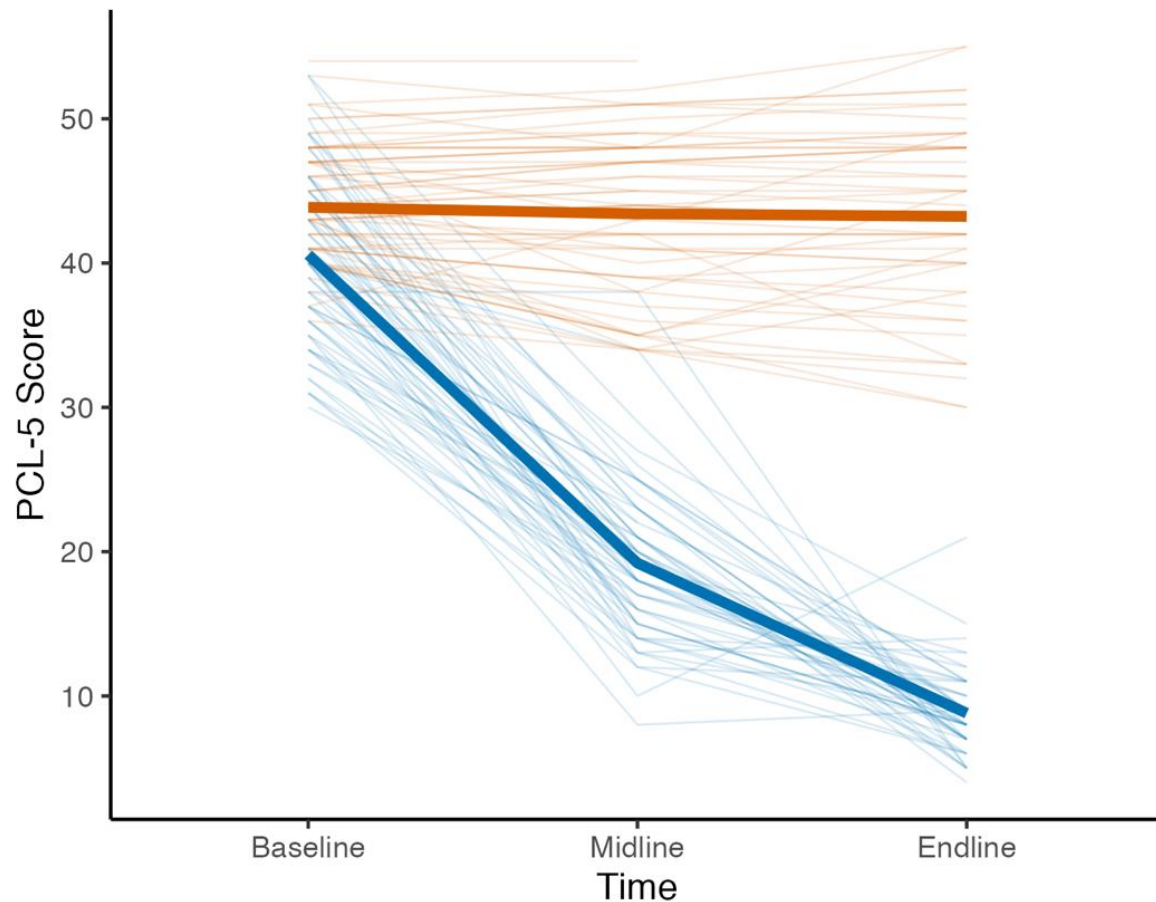
The treatment group had lower symptom scores than the control group.

Secondary Figures

The figures in this section serve an exploratory function. They offer supplementary insights that enrich the primary findings.

The figures include:

- i. Spaghetti plot to show individual participant change over time
- ii. UpSet plot to visualize the overlap between PTSD and its comorbidities



Thin lines = individuals. Thick lines = group means.

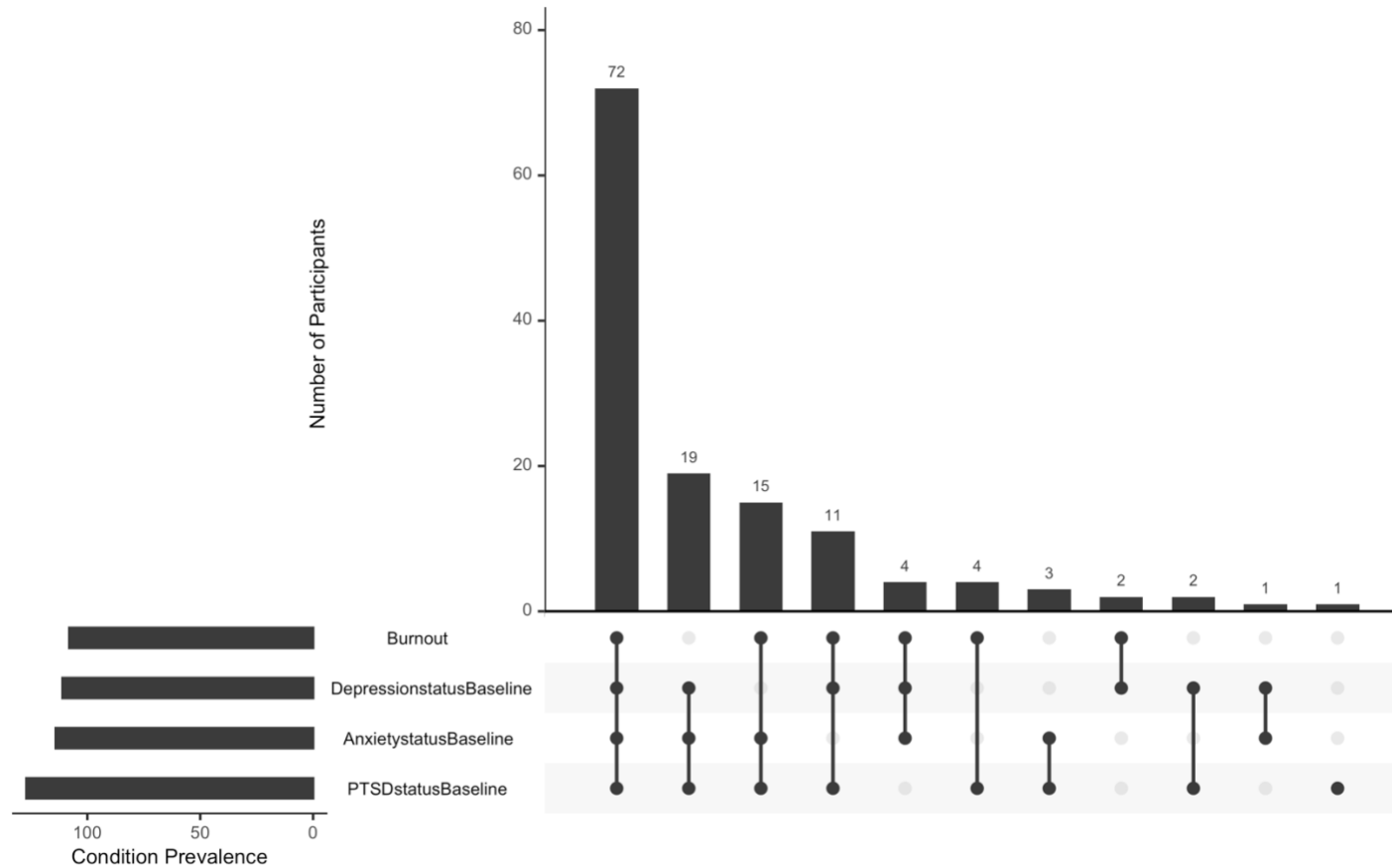
Spaghetti Plot

This figure shows individual participant trajectories over time. It checks whether improvement is consistent across individuals.

The plot was desirable for its ability to reveal individual trajectories, which complement group summaries. A mean-only line graph would mask individual variability while a slope chart would only capture two time points and ignore midline data.

Most treatment participants show downward trajectories. This confirms that the effect is not driven by outliers.

Therefore, the majority of individuals improved.



UpSet Plot (Comorbidity)

The chart displays the overlap between PTSD, anxiety, depression, and burnout. It demonstrates the complexity of mental health burden.

The plot was desirable for its ability to scale well and show intersection sizes. A Venn diagram would fail to scale well and become unreadable after three sets while a correlation matrix would fail to show overlap patterns.

Many participants experience multiple co-occurring conditions, highlighting a need for holistic interventions like EMDR.

Therefore, mental health conditions frequently co-occur in this population.